

Algorithms, pseudocode, programs

Intro. to Computing

<https://mandarm.github.io/computing.html>

What is an algorithm?

An algorithm is a step-by-step procedure to accomplish some objective.

Examples

- How to withdraw money from an ATM.
- How to play the Indian National Anthem on a computer.
- How to obtain the distribution of blood groups in this class.
- How to compute the square root of a positive integer.

Properties

- Typically written in natural language, e.g., English
- Must be *sufficiently detailed and unambiguous* for the intended person

What is pseudocode?

Also a step-by-step procedure to accomplish some objective, but looks somewhat like a computer program

Properties

- Preferably limited to *primitives* (simple, elementary actions)
- Uses some standard programming constructs in a language-independent way
 - *conditionals*: check whether a condition holds in order to determine what to do
 - *loops*: do something repeatedly a fixed number of times, or until some criteria are fulfilled
- **BUT** no strict rules regarding the format
- Must be *sufficiently detailed and unambiguous* for the intended person

Examples: problem 1

Problem 1

Consider a reservoir, fitted with a set of taps and drains. You are given the capacity of the reservoir, and details about the taps and drains attached to the reservoir. Determine whether a partially full reservoir ever becomes empty or full.

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Input:

- L capacity of the reservoir in litres
- T number of taps attached to the reservoir
- D number of drains attached to the reservoir
- t_i time in minutes that it takes to fill an initially empty reservoir if all drains stopped, and only the i -th tap is opened ($1 \leq i \leq T$)
- d_j time in minutes that it takes to drain an initially full reservoir if all taps shut off, and only the j -th drain is opened ($1 \leq j \leq D$)

Examples: problem 1

Pseudocode

- I Initially, set variables X and Y to 0.
- II **for** i in $\{1, 2, 3, \dots, T\}$
 $X \leftarrow X + \underline{\hspace{2cm}}$.
- III **for** j in $\{1, 2, 3, \dots, D\}$
 $Y \leftarrow Y + \underline{\hspace{2cm}}$.
- IV **If** $X \underline{\hspace{1cm}} Y$
 then print "*A partially empty reservoir never fills up if all taps and drains are kept open.*"
 else print "*An initially empty reservoir fills up in $\underline{\hspace{2cm}}$ minutes if all taps and drains are kept open.*"
- V **If** $X \underline{\hspace{1cm}} Y$
 then print "*A partially full reservoir never becomes empty if all taps and drains are kept open.*"
 else print "*An initially full reservoir becomes empty in $\underline{\hspace{2cm}}$ minutes if all taps and drains are kept open.*"

Examples: problem 2

Problem 3

A merchant buys impure salt from N suppliers $S_1, S_2, \dots, S_N$, refines it, and sells the purified salt (all impurities are extracted and discarded) at Rs. c/kg . Fill in the blanks (marked as (i)-(v)) appropriately in the algorithm given below, so that it determines the value of c for which this entire transaction results in a total profit of Rs. X for the merchant.

Inputs: Three arrays $P[\]$, $Q[\]$, and $U[\]$, each containing N elements, indexed from 1 to N .

- $P[i]$ denotes the purity of the salt bought from S_i . It is a number between 0 and 1, with 1 denoting completely pure salt.
- $Q[i]$ denotes the weight (in kg) of impure salt purchased from S_i .
- $U[i]$ denotes the cost price (per kg) of impure salt purchased from S_i .

Examples: problem 2

Pseudocode

- I Initialise $CostPrice \leftarrow 0$, $Weight \leftarrow 0$, $i \leftarrow 1$.
- II While _____ (i)
 - (a) $CostPrice \leftarrow CostPrice +$ _____ (ii)
 - (b) $Weight \leftarrow Weight +$ _____ (iii)
 - (c) $i \leftarrow$ _____ (iv)
- III Output $c =$ _____ (v) $\leftarrow$ This is often called the *return value*.

Examples: problem 3

Problem 3

Compute the GCD of two integers a and b .

Pseudocode

- I Let r be the remainder when a is divided b .
- II If r is 0, required GCD (or *return value*) is b .
- III Otherwise?

Corner case: a or b can be 0 or 1

What do we need?

- **Variables:** to store values, and to use them in calculations
 - variables have (dynamic) types
- **Operators** (and functions):
 - arithmetic operators
 - relational operators
 - logical operators
 - string operators / functions
- **Statements:** executing actions
 - assignment statements
 - conditional statements
 - loops
- **Input and output methods**
- **Built in structures:** lists, dictionaries, ...